



Assignment 6

1 Instructions

- This exercise will be submitted together with next exercise, as latest, **on Tuesday, April 2nd, at midday**.
- **Before implementing**, draw **UML Class Diagrams**, save them as images and **copy each one into the root of their project**.
- Rename project. Export them to a **single** archive file `surname1_surname2_name_lab07`.
- Fill the **checklist** and copy to the root of the project.

2 Exercise 1: LinkedList and ArrayList

You must develop two different implementations of the List interface provided: a `LinkedList` class and an `ArrayList` class. For doing so, read carefully the documentation of the `Collection` and `List` interfaces, since they specify the contract for your lists, the exact behaviour every operation must exhibit.

2.1 Unit Tests

Test all their public methods. As always, remember that your tests must be as comprehensive as possible. There are many different scenarios that should be tested for many of their operations: when an element is at the beginning, in the middle or at the end; when the list is empty; when null elements are involved; etcetera. Think carefully about all of them.

You should also think about the best way of testing your lists.

On one hand the test functionality is defined by the `List` interface. But unit tests test concrete implementations, not interfaces, obviously.

So, how could we avoid code duplication in our test classes (as the behaviour is the same, the tests should also be exactly the same) while at the same time testing all of that functionality for both list implementations?

3 Using your lists in the project DOME

Once you finish the implementation of both lists and all their tests, import your last version of DOME (including its tests).

Now, replace every reference to the Java list with your own implementations.

Try to use both implementations of list in your project (for example, one type of list for all the items of the database and the other one for the borrowable items).

Run DOME tests again.